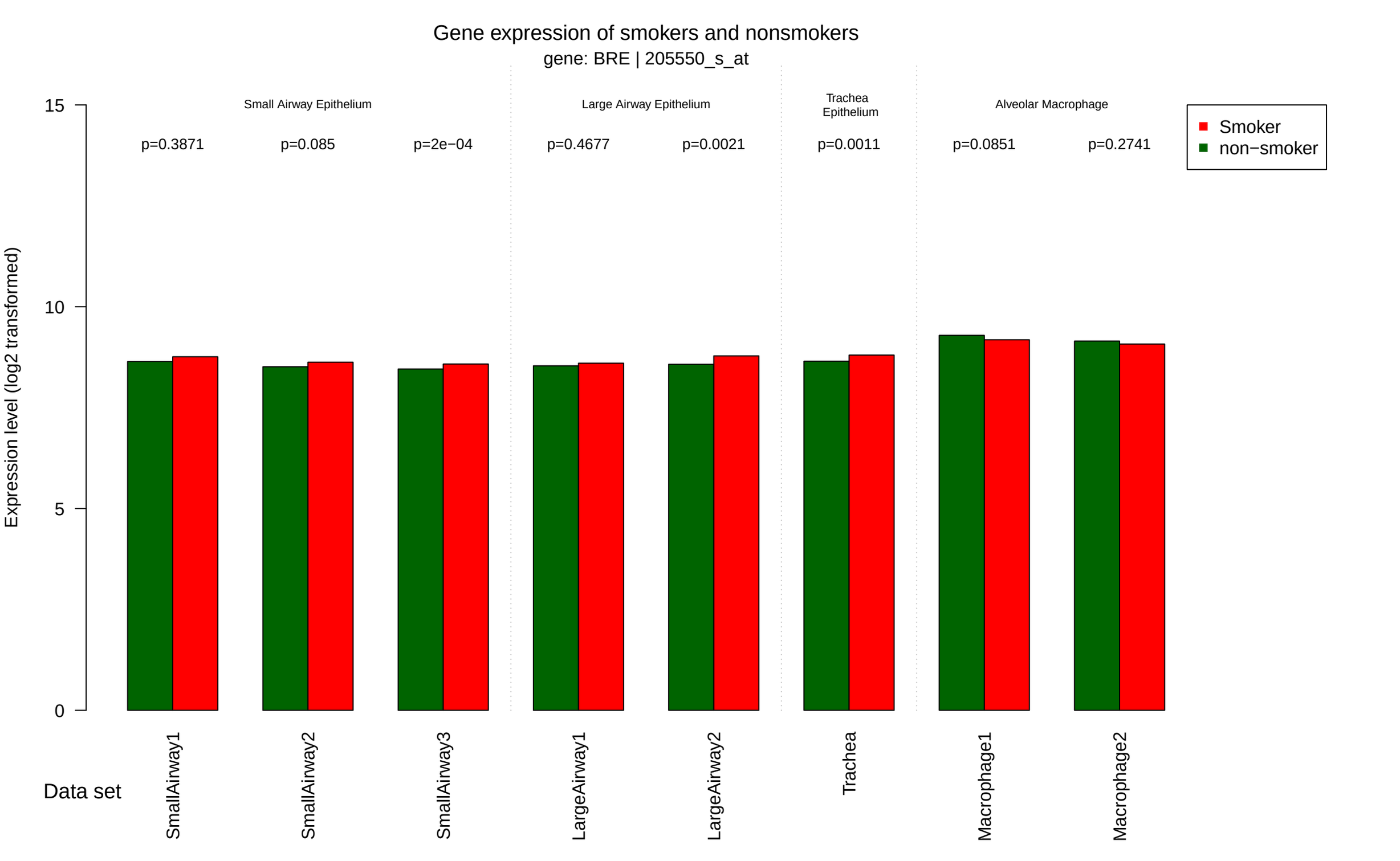
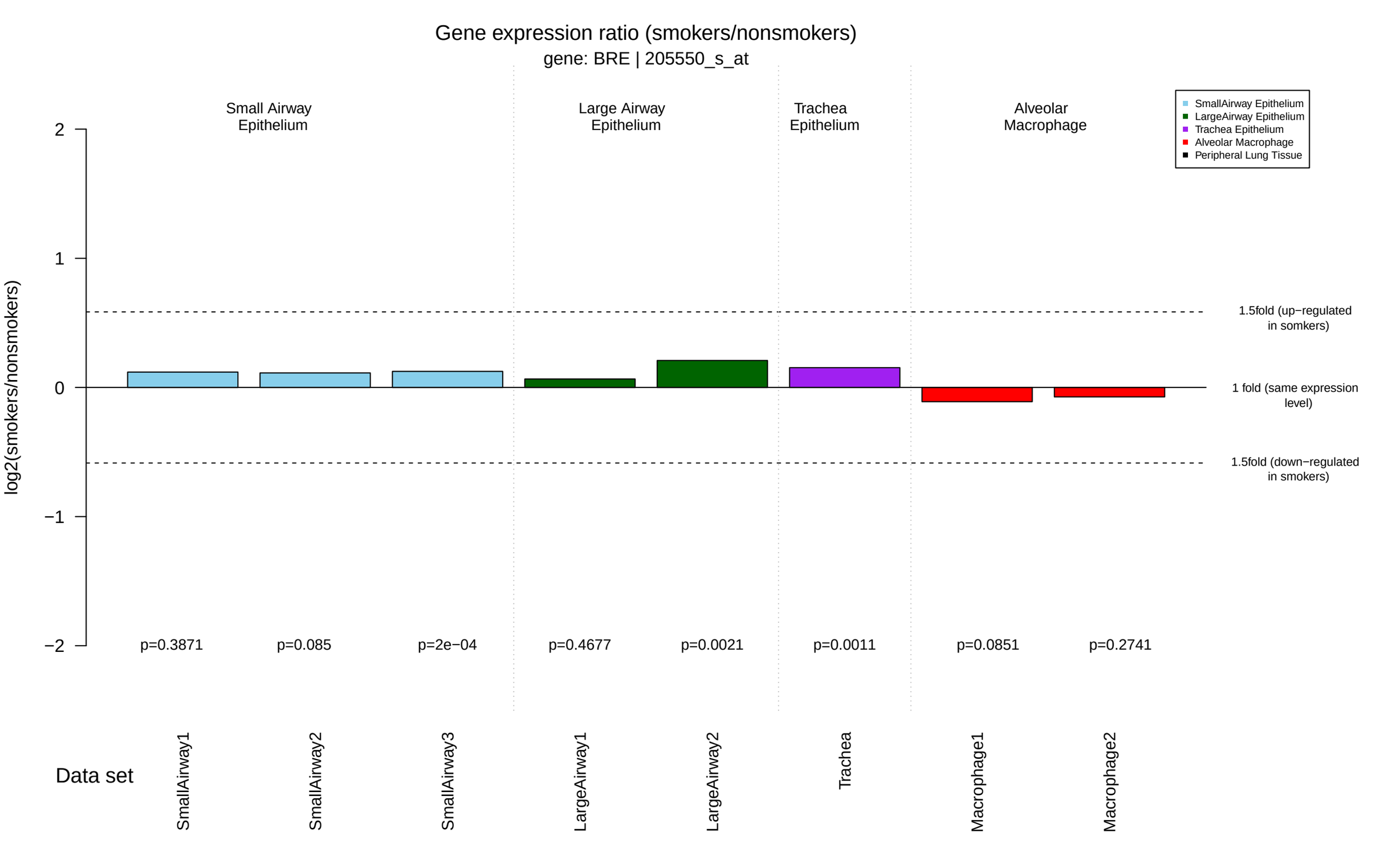
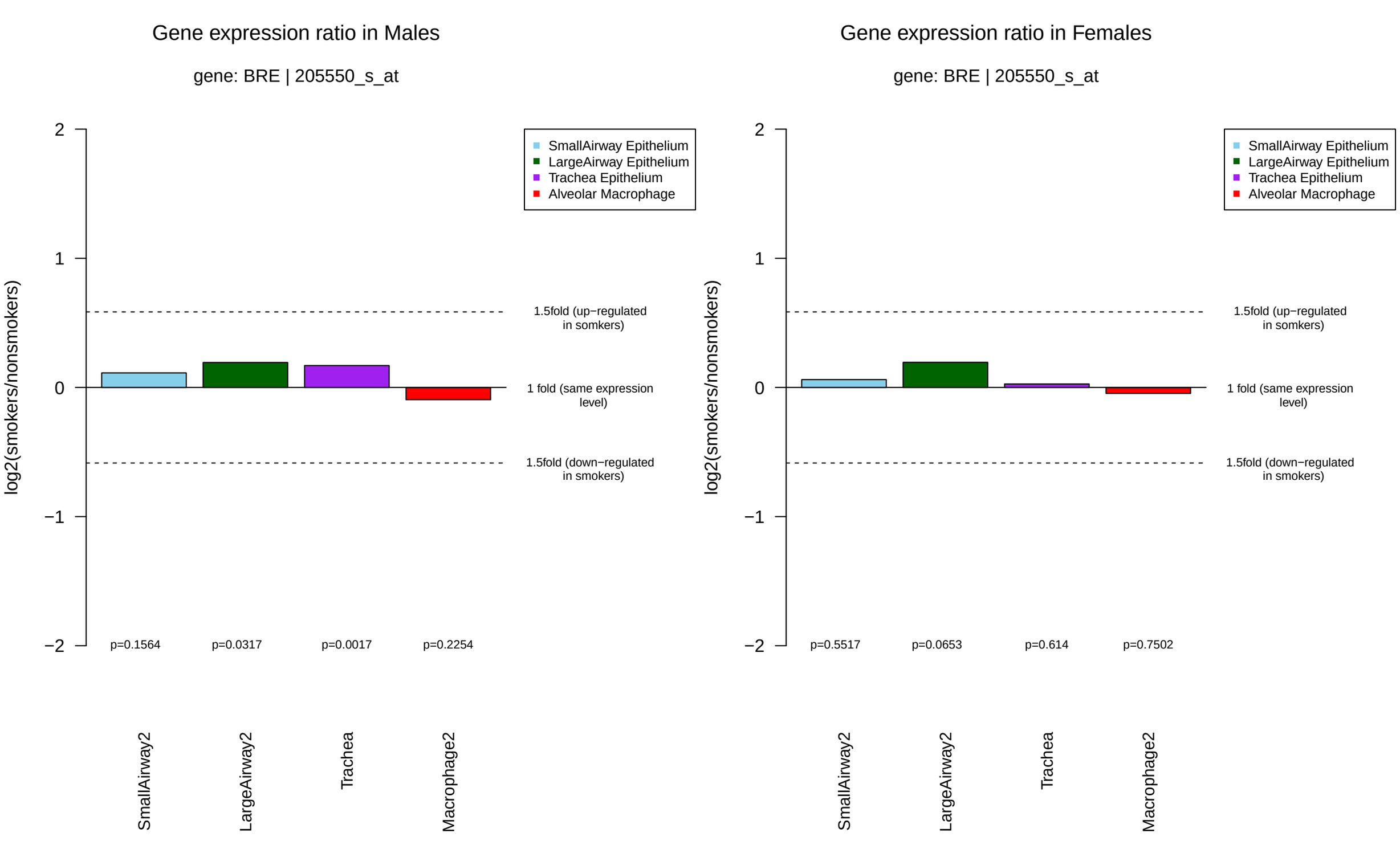
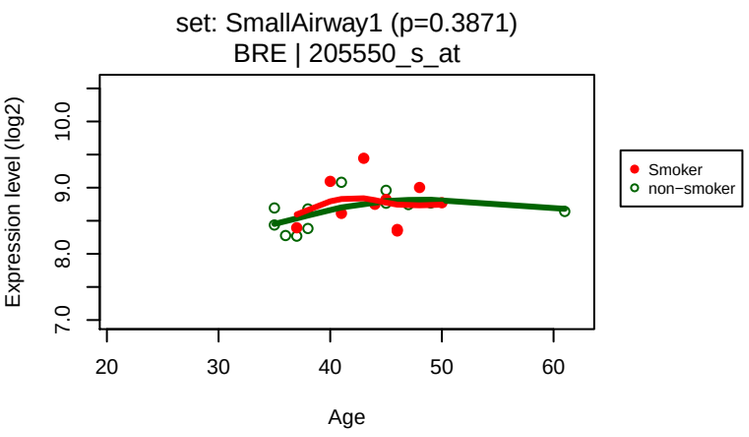
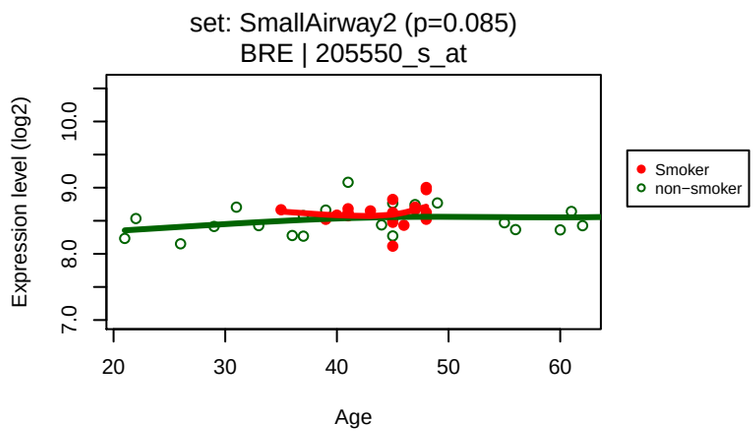
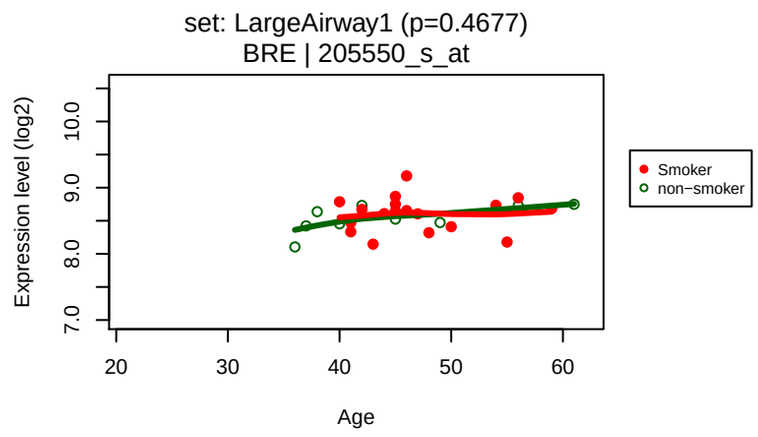

Gene expression of smokers and nonsmokers
gene: BRE | 205550_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3871
p=0.085
p=2e−04
p=0.4677
p=0.0021
p=0.0011
p=0.0851
p=0.2741
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

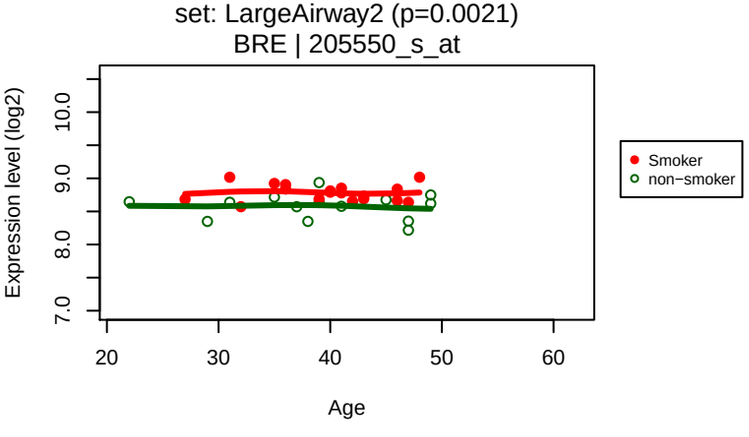

Gene expression ratio (smokers/nonsmokers)
gene: BRE | 205550_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3871
p=0.085
p=2e−04
p=0.4677
p=0.0021
p=0.0011
p=0.0851
p=0.2741
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

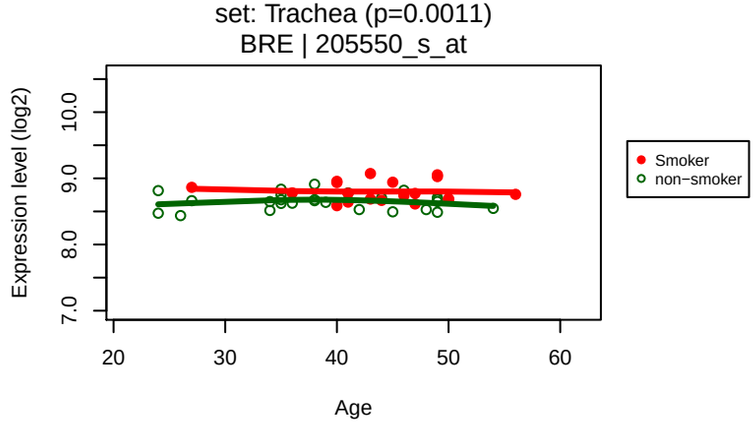

Gene expression ratio in Males
gene: BRE | 205550_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1564
p=0.0317
p=0.0017
p=0.2254
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRE | 205550_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5517
p=0.0653
p=0.614
p=0.7502
SmallAirway2
LargeAirway2
Trachea
Macrophage2

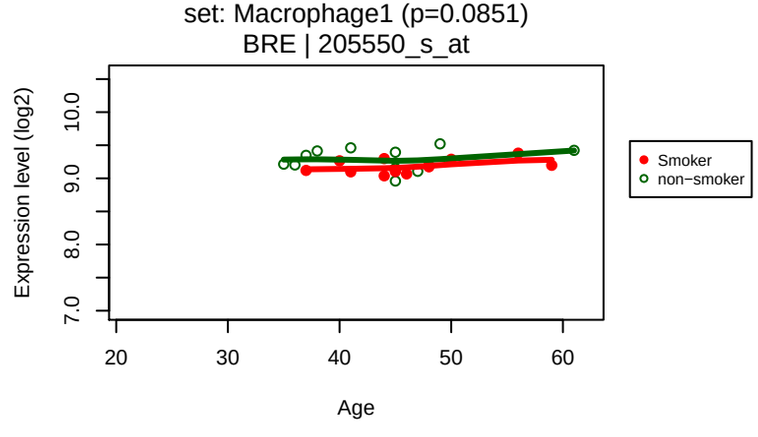

set: SmallAirway1 (p=0.3871)
BRE | 205550_s_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

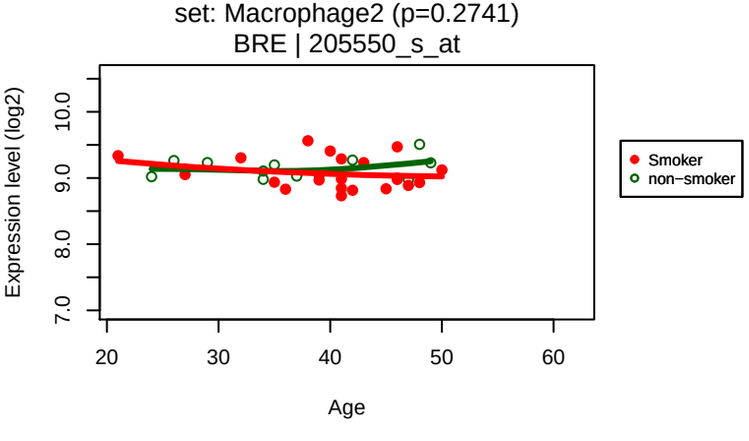

set: SmallAirway2 (p=0.085)
BRE | 205550_s_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

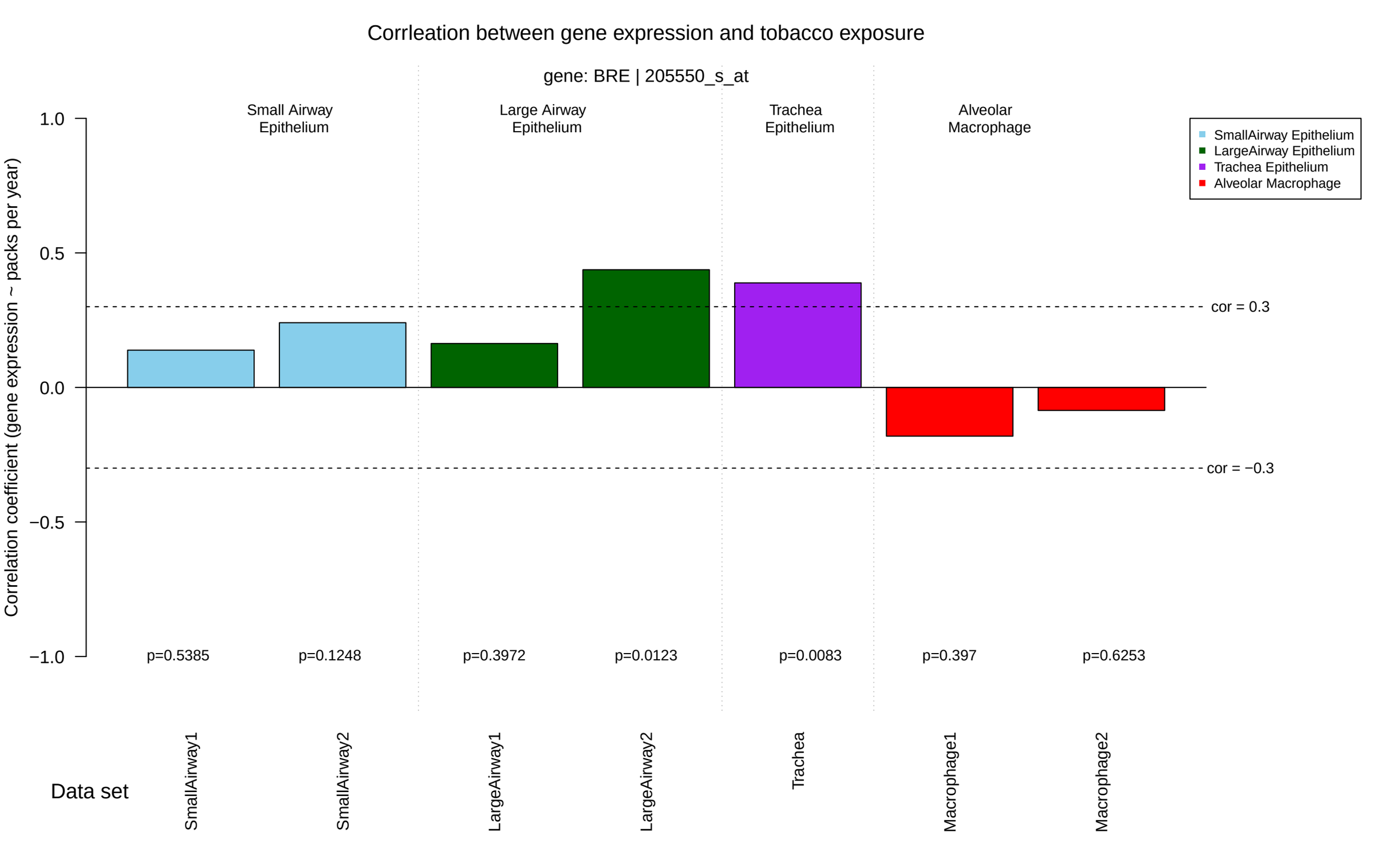

set: LargeAirway1 (p=0.4677)
BRE | 205550_s_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0021)
BRE | 205550_s_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
BRE | 205550_s_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0851)
BRE | 205550_s_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2741)
BRE | 205550_s_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRE | 205550_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5385
p=0.1248
p=0.3972
p=0.0123
p=0.0083
p=0.397
p=0.6253
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2